

PRESENTED BY THE ENGINEERING GUILD

FIELD BRIEFING // v1.0



STEM ENGINEERING CHALLENGE SERIES

OPERATION SKYLINE

The Rescue Glider Challenge

Salvage. Engineer. Deploy. Save the zone.

Saturday, 15 August 2026 • Innovation Hangar, Main Auditorium

Disaster Strikes Again.

● MISSION BRIEFING

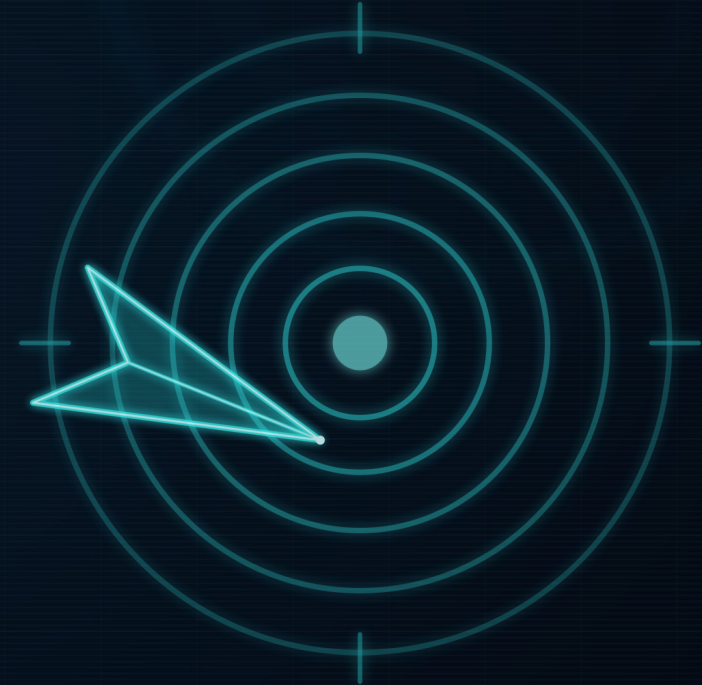
Disaster strikes again.

While approaching the rescue zone, teams hit dangerous terrain and severe conditions that halt all ground movement.

The rescue point lies beyond a mountain ridge — and time is running out.

Using only salvaged aircraft materials and available resources, each team must design and build an emergency glider capable of carrying critical supplies across the obstacle.

The mission now depends on your engineering creativity, precision, and teamwork.



Your Objective, Rescue Engineers

Build a flight-worthy glider from salvage that clears the ridge and delivers its payload to the rescue zone.

01

Engineer the Airframe

Design & assemble a glider from the salvaged materials on your bench.

02

Secure the Payload

Carry the supply capsule — mass counts, but it must arrive intact.

03

Clear the Ridge

Achieve the distance to cross the obstacle and reach the far side.

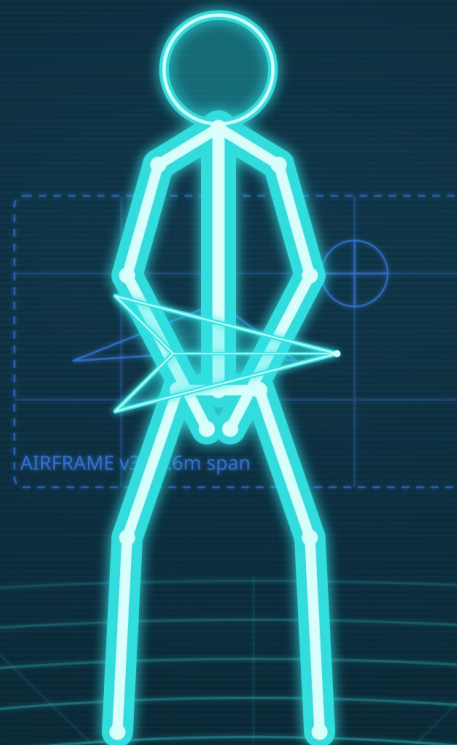
04

Hit the Zone

Land on-target. Accuracy over the rescue point wins the mission.

SEQUENCE 01 / 03

ASSEMBLE THE GLIDER



BUILD

Salvage becomes airframe. Hands, blueprints, and precision under pressure.



04 / 10 BUILD

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SEQUENCE 02 / 03

DEPLOY



LAUNCH

Wind up. Read the air. One clean release sends the payload skyward.



05 / 10 LAUNCH

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SEQUENCE 03 / 03

TARGET: RESCUE ZONE



IMPACT



The arc holds true. Supplies strike the zone. Mission accomplished.



Field Regulations

01

Salvage Only

- Balsa, paper, card, tape & string from the supply crate
- No pre-built kits or powered components
- One roll of tape per team — spend it wisely

02

Build Window

- 45 minutes to design & construct
- 10-minute strategy huddle before the clock
- Tools down when the siren sounds

03

Launch Rules

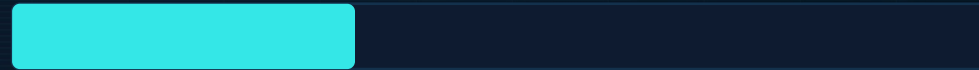
- Hand-launch only, from behind the deploy line
- One practice throw, three scored attempts
- Glider must be intact after each flight

How the Mission Is Scored

35%

Distance

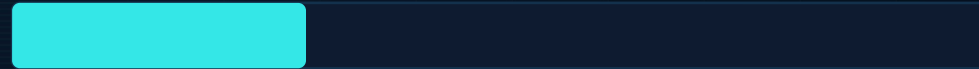
Clearing the ridge — measured from the deploy line to touchdown.



30%

Target Accuracy

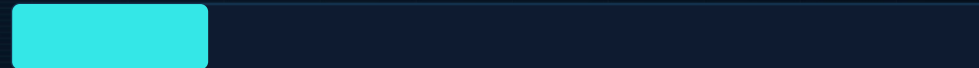
Proximity of landing to the marked rescue zone.



20%

Design & Craft

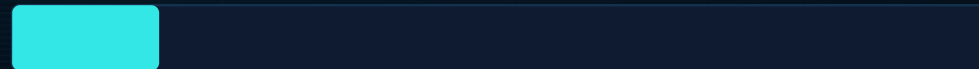
Structural creativity, aerodynamics, and build quality.



15%

Teamwork

Coordination, safety, and clean execution under time.



Schedule of Operations

09:00



Check-in & Briefing

Teams assemble. Mission parameters issued.

09:30



Salvage Distribution

Supply crates handed out. Inspect your materials.

09:45



Build Window Opens

45 minutes on the clock. Engineer the airframe.

10:30



Deploy & Fly

Hand-launch trials across the ridge.

11:30



Judging & Scoring

Distance, accuracy, design, and teamwork tallied.

12:15



Awards Ceremony

Rescue honors awarded. Mission debrief.



D E P L O Y

Good Luck, Rescue Engineers

The ridge is waiting. The zone is counting on you.

Salvage. Engineer. Deploy.

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